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Stat 50

HW # 20
Section 5.7

(7)

(a) 23.337

(b) 4.404

(c) 16.750

(d) 0.4012

(e) 30.813

(f) 14.041

(22) $n = 25$, $S = 15$, 95%.

For $\sigma^2 = \left(\frac{(15)^2 24}{39.364} , \frac{24(15)^2}{12.401} \right)$

$\sigma^2 = (137.18 , 435.45)$

We are 95% confidence that the true variance of population is within 137.18 and 435.45. If we repeated this process a large number of time with 25 sample size and 95% confidence Interval.

(5) $n = 18 \qquad S = 8 \qquad 95\%$

$$\sigma^2 : \left(\frac{17(8)^2}{30.191} , \frac{17(8)^2}{7.564} \right)$$

$$= (36.04 , 143.84)$$

We are 95% confidence that true varience of population is with in the interval (36.04, 143.84). If we repeated this process a large number of time with 18 Sample size and 95% C.I's

(T) $\mu = 14 , \quad n = 12$

(a) $$S = \sqrt{\frac{(14 - 14.02)^2 + (13.97)^2 + \dots + (14.12 - 14)^2}{11}}$$

$$= \sqrt{0.00377}$$

$$S = 0.0614$$

(b) 98%

$$\sigma : \left(\sqrt{\frac{11(0.0614)^2}{24.725}} , \sqrt{\frac{11(0.0614)^2}{3.053}} \right)$$

$$= (0.04095 , 0.117)$$

We are 98% confidence that the true Standard deviation is with in the interval of (0.041, 0.117). If we repeated this process a large number of times and 95% CI's.

(8)

$n = 6$, $\bar{x} = 20.383$, 99%.

$$S = \sqrt{\frac{(19.1 - 20.383)^2 + (20.8 - 20.383)^2 + \cdots + (19.7 - 20.383)^2}{n-1}}$$

$$S = 0.85766$$

$$\sigma^2 : \left(\frac{5(0.83766)^2}{16.750} , \frac{5(0.83766)^2}{0.412} \right)$$

$$\sigma^2 : = (0.20945 , 8.5154)$$

We are 99% confidance that the true variance is with in the interval of (0.20945, 8.5154) If we repeated this process a large number of time and 99% CI's.

(9) $n = 12$, 95 %.

$\bar{x} = 3.1028$ $S = 0.18053$

$$\sigma^2: \left(\frac{12(0.18053)^2}{21.920} , \frac{11(0.18053)^2}{3.816} \right)$$

$$\sigma^2: = (0.01636 , 0.09395)$$

We are 95 % confidence that the true Variance of populations within the interval of (0.01636 and 0.09395) If we repeated this process a large number of times and 95 % CI's.